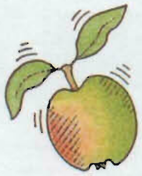
GRAVITY



WITHOUT GRAVITY, we would fly off the spinning Earth and into space. Gravity is a force of attraction that acts between any two objects. The objects can be as large as galaxies or as small as subatomic particles. The strength of the gravity between two objects depends on their masses and the distance between them. Objects with large masses exert a strong force of gravity. Objects far apart attract each other weakly.

Centre of gravity

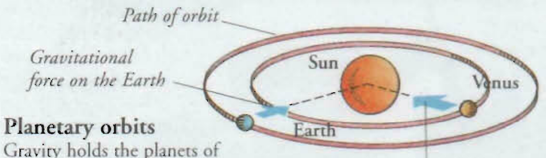
Centre of gravity is directly below the string, making the object very stable.

Every object consists of tiny particles of matter. Each of these particles has a small force of gravity acting upon it. Together, the forces act like a single force pulling downwards at just one point, called the centre of gravity. An object will balance when it is supported in line with its centre of gravity. Balancing is easiest if the object has a low centre of gravity.

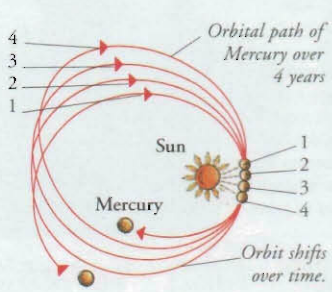
Gravity in space

Gravity is a universal force, because it acts between any two objects, wherever they are in the Universe. The force that keeps our feet firmly on the ground is the same one that holds huge clusters of stars together as galaxies.

Galaxies
A typical galaxy is about 100,000 light-years across. The stars are so massive that gravity can still act over this huge distance, preventing the stars from drifting off into space.



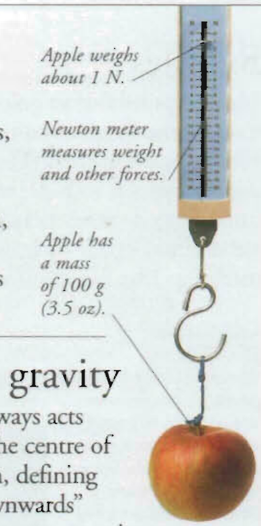
Planetary orbits
Gravity holds the planets of the Solar System in orbit around the Sun. Venus and the Earth have similar masses, but because Venus is closer to the Sun than the Earth, the force of gravity keeping it in orbit is greater.

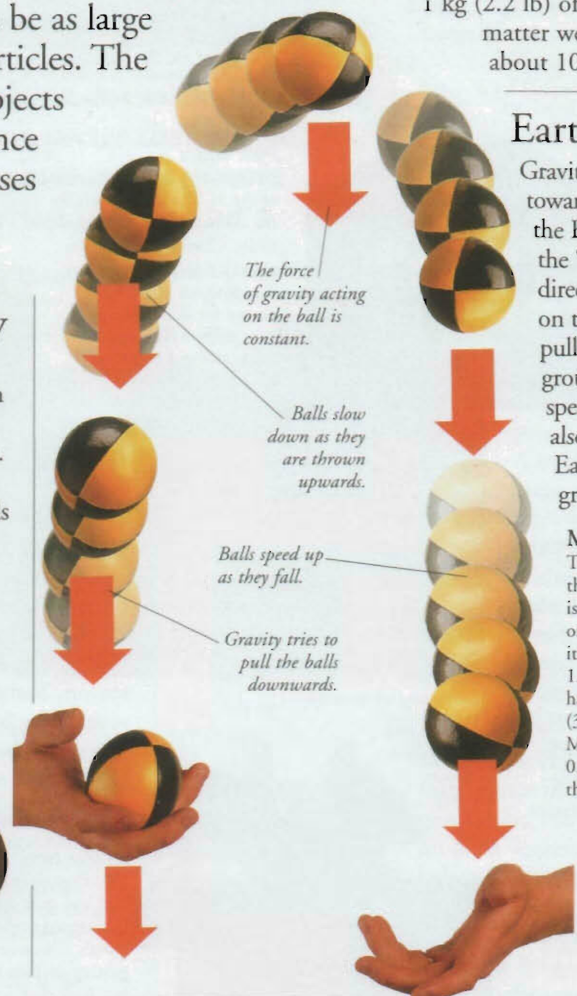


General Relativity
In 1915, German-born physicist Albert Einstein published his Theory of General Relativity. This theory sees gravity not as a force, but as a curvature of space caused by bodies of matter. In 1919, the theory was used successfully to explain why Mercury's orbit gradually varies over time.

Weight

The force of gravity acting on an object is called weight. Like all forces, weight is expressed in units called newtons (N). An object's weight is directly related to its mass. On Earth, 1 kg (2.2 lb) of matter weighs about 10 N.

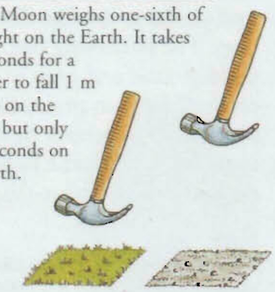




Earth's gravity

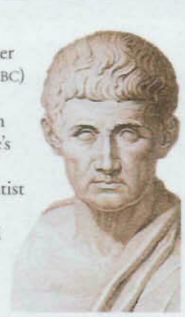
Gravity always acts towards the centre of the Earth, defining the "downwards" direction at every point on the planet's surface. Gravity pulls a juggling ball towards the ground, slowing it as it rises, and speeding it up as it falls. The ball also pulls on the Earth, but the Earth is so massive that the ball's gravity has no noticeable effect.

Moon's gravity
The Moon is smaller and has less mass than the Earth, so the force of gravity is weaker on the Moon. A hammer on the Moon weighs one-sixth of its weight on the Earth. It takes 1.1 seconds for a hammer to fall 1 m (3.3 ft) on the Moon, but only 0.44 seconds on the Earth.



Aristotle

The Greek philosopher Aristotle (c.384–322 BC) believed that heavy objects fall faster than lighter ones. Aristotle's ideas were accepted until the Italian scientist Galileo Galilei (1564–1642) showed that gravity pulls all objects to Earth at the same speed.



Harbour at low tide

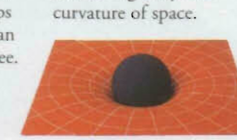
Tides
Twice each day, the waters of the ocean rise a little and then fall back. This movement is called a tide, and it is caused by the pull of the Moon's gravity. The Sun also influences tides. When the Earth, Sun, and Moon are in line, their combined gravity produces tides that are higher than normal, called spring tides.

Timeline

4th century BC Aristotle proposes that stones fall to the ground simply because they are heavy, and that smoke rises because it is light.

1604 Italian scientist Galileo Galilei investigates how objects fall to Earth.

17th century English physicist Isaac Newton publishes his Law of Gravitation, perhaps inspired by seeing an apple fall from a tree.



Model showing how space curves around a planet.

1915 Einstein's Theory of General Relativity describes gravity as a curvature of space.

1919 English astronomer Arthur Eddington (1882–1944) obtains proof of Einstein's theory by observing light, reaching Earth from a distant star, being bent by the Sun's gravity.

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